DSA ki Baithak

CODE BAITHAK

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Recap (Previous Lecture Connection):

* Intro to Programming
* Flowcharts
* Pseudocodes

How do we talk to a Computer ?

What is a Programming Language ?

* It is a way to give instructions to a computer.
* It acts as a bridge between human and computer.

<u>Example</u> : Java, C, Python

Human readable Code → Machine understandable instructions

Types of Programming Languages

① Low Level Language

* Machine friendly
* Hard to write
* Fast execution

Ex: Machine Language (0 & 1)
Assembly Language

② High Level Language

* User friendly (Human Readable)
* Easy to write
* Slightly slower

Ex: Java, Python
C, C++

But Computer Understands Only 0 and 1

* CPU understands only binary, i.e, 0 & 1
* So, how does High Level Code run?

Using :

1. Compiler
2. Interpreter

Compiler vs Interpreter

Compiler

* converts entire code at once
* Generates executable file (.exe)
* Faster Execution

Ex: C, C++

Whole program → Machine code

Interpreter

* Converts line by line
* No separate executable file
* Slower but easy debugging

Ex: Python

Line by line → Execution

Where does Java fit ?

* Java is a High Level Language.
* Java is both Compiled and Interpreted.

How Java works Internally ?

Steps :

1. .java file → compiled by Java Compiler (javac)
2. Generates Bytecode (.class file)
3. JVM converts Bytecode → Machine Code

Flow Diagram :

.java file → javac → .class file → JVM → Machine Code → Output

(Bytecode)

System Independent

Why Java is Platform Independent ?

* Because of Bytecode, which is platform independent.
* JVM runs on every OS.

(JVM is platform dependent)

"Write Once, Run Anywhere"

Bytecode is same for Windows, Mac, Linux but each has its own JVM

C++ vs Java

```
#include <iostream>
using namespace std;
int main(){
    cout << " Hello";
}
```

.cpp file
(source code)

Compiler
(entire file)

011010 ...
100101 ...
- - - - - - - - -
- - - - - - - - - -

.exe file
(Machine code)

Run

Output

```
import java.util.*;
public class Main {
    public static void main(){
        System.out.print("Hello");
    }
}
```

.java file
(source code)

Compiler
(entire file)

.class file
(Intermediate/ Byte code)

Interpreter
(line by line)

101011 ...
00010 ...
- - - - - - - - -
- - - - - - - - - -

.exe file
(Machine code)

Run

Output

JDK vs JRE vs JVM

JVM (Java Virtual Machine)

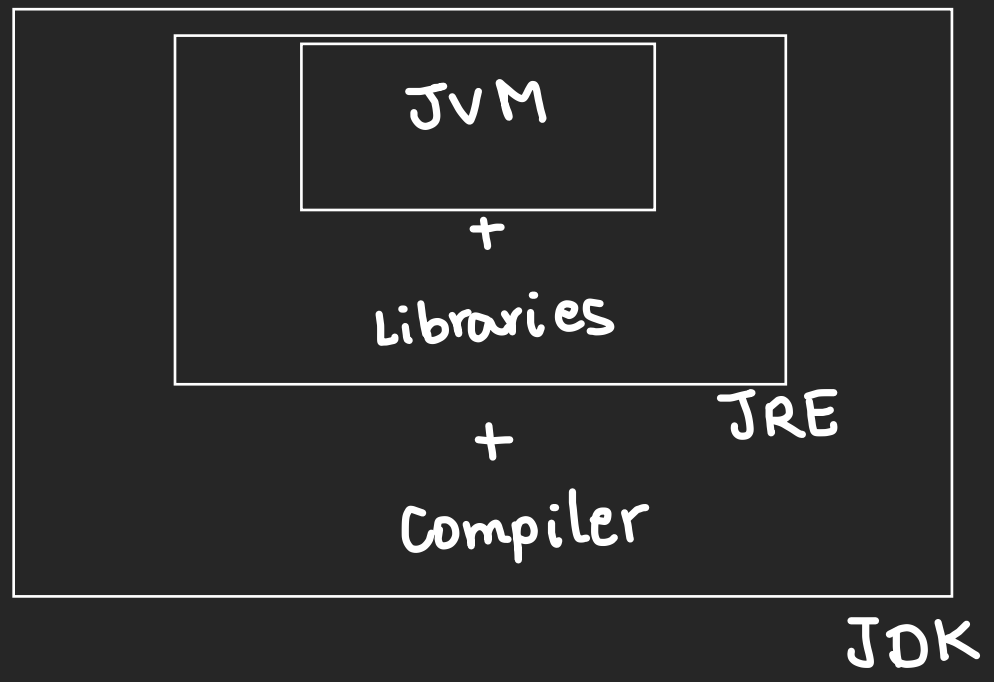
* Runs Java Bytecode
* Converts Bytecode → Machine Code

JRE (Java Runtime Environment)

* Provides JVM + Required Libraries
* Used to run Java Programs

JDK (Java Development Kit)

* Contains JRE + Compiler (javac)
* Used to develop Java Programs



Advantages of Java

* Platform Independent
* Object Oriented
* Secure & Robust
* Large community support
* Used in Android, Backend, Enterprise Apps